SUBJECT: Microprocessor and Computer Architecture

Semester: III

		1 Mark Each
	Module	**Question**
1.	1	Define RAM and its usage.
2.	1	List the features of 8086 microprocessor
3.	1	Define Computer Architecture
4.	1	Define Computer Organization
5.	1	Draw Von Neumann Model of processor design
6.	1	Define Performance Measure of Computer Architecture.
7.	1	List the components of 8086 Architecture.
8.	1	List the Classification of Instruction Set
9.	1	List the Addressing Modes of 8086
10.	1	Define Addressing Modes
11.	1	Evaluate the ADD AL,BL instruction for AL=20H and BL=28H, and state the status of CF,AF,ZF and PF
12.	1	Explain the register indirect addressing mode with example
13.	1	Explain the PUSH instruction
14.	1	List 4 data transfer instruction in different addressing modes.
15.	2	Define stages in 3 step Instruction Cycle
16.	2	Describe parts of 2 address instruction
17.	2	Define Interrupt
18.	2	Define Instruction Pipelining concept.
19.	2	What is pipeline concept
20.	2	Define Hard wired control unit
21.	2	Define Micro-operation
22.	2	Explain the usage of Micro-operation
23.	2	Define Micro-code
24.	2	what is operand data types
25.	2	Define term softwired control Unit
26.	2	Define term hardwired control unit
27.	3	Define Port pins of 8086
28.	3	Define are the segment address and offset?
29.	3	Discuss use of RESET pin in 8086
30.	3	what is min and max mode of 8086.
31.	3	Define the memory banks of 6086
32.	3	Define Memory Segmentation
33.	3	Explain the use of clock signal.
34.	3	What is the role of ALE in timing diagram

35.	3	list the signals/pins used in read operation.
36.	3	Define pins/signals used in Maximum mode
37.	3	Define the pin and its status used to select maximum mode in 8086
38.	3	Define the pin and its status used to select minimum mode in 8087

		5 Mark Each
	Module	**Question**
1.	1	What are the fundamental differences between computer organization and computer architecture?
2.	1	How does the Central Processing Unit (CPU) work? What are registers, ALUs (Arithmetic Logic Units), and control units?
3.	1	Differentiate between computer organization and computer architecture
4.	1	Describe Von Neumann Model.
5.	1	Extract Performance Measure of Computer Architecture.
6.	1	Explain the string related instructions of 8086.
7.	1	Write an ALP for 8086 to perform Multiplication operation for any two 8 bit number.
8.	1	Explain Addressing Modes.
9.	1	Define Indirect Addressing modes of 8086 with example.
10.	1	Implement and ALP for 8086 to add two 8 bit number.
11.	1	Explain the operation of Indirect addressing and register addressing mode.
12.	1	Describe the following instructions: TEST, CMP, SAR,ROR and CALL related to 8086.
13.	1	Draw and explain architecture of 8086.
14.	2	Explain Flynn's Classifications.
15.	2	Describe Parallel Processing Concepts.
16.	2	Explain Microinstruction Sequencing and Execution.
17.	2	Explain the 3- step Instruction Cycle.
18.	2	Discuss the fundamental concepts of parallel processing.
19.	2	Explain different instruction formats with suitable example.
20.	2	Write a short note on micro-instruction sequencing and execution.
21.	2	Describe in detail the organization of a typical CPU.
22.	2	Explain the Hardwired control unit.
23.	2	Explain the concept of instruction pipelining.
24.	2	Define instruction pipelining and demonstrate its usage.
25.	2	Explain Microprogrammed Control Unit.
26.	2	Differentiate between hardwired and micro-programmed control unit.
27.	3	Explain flag register bits of 8086.
28.	3	Explain programming model of 8086.
29.	3	Discuss the 8086 microprocessor system clock signal and T states.

30.	3	Write a short note on advantages of memory segmentation in 8086.
31.	3	Explain memory banks for 8086 processor.
32.	3	Draw the pin diagram of 8086 microprocessor.
33.	3	Explain minimum mode configuration of 8086.
34.	3	Explain maximum mode configuration of 8086.
35.	3	Differentiate between minimum and maximum mode of 8086.
36.	3	Draw and explain Memory Read operation timing diagram for Minimum mode.
37.	3	Draw and explain I/O write operation timing diagram for minimum mode.
38.	3	Draw and explain write operation timing diagram for minimum mode.